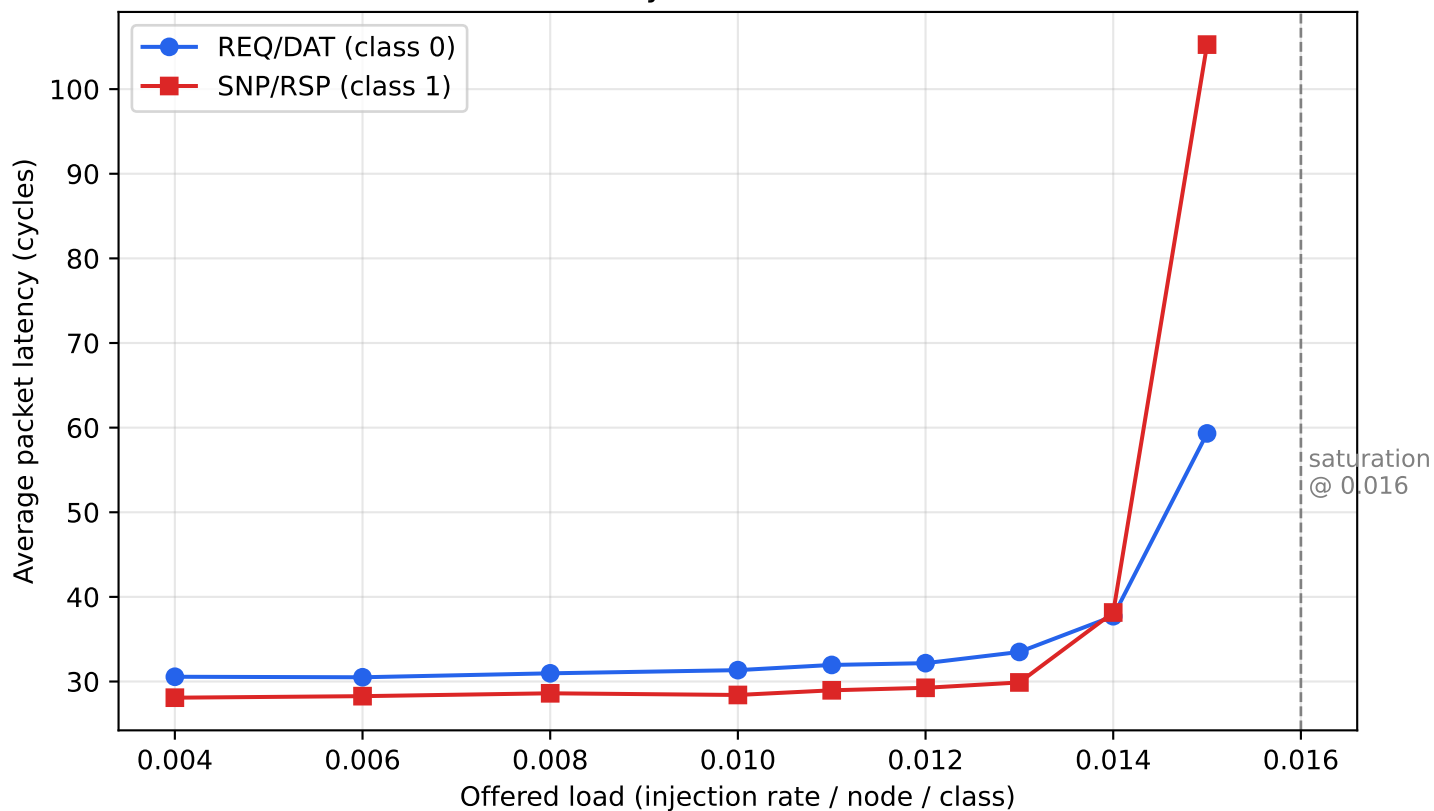
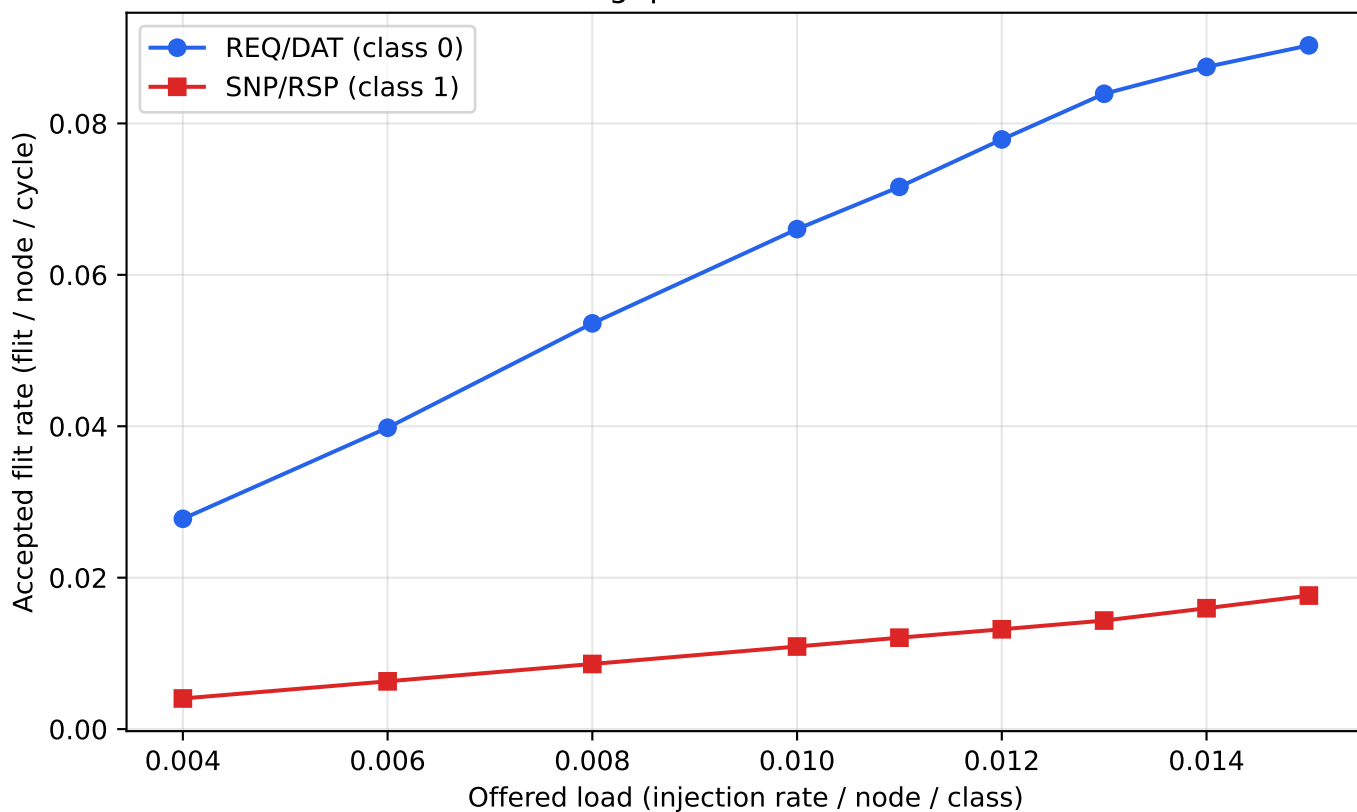


6x7 Mesh + CHI 4 Independent Physical Channels Injection-Rate Sweep (snoop_factor = 1.0)

Latency vs. Offered Load



Throughput vs. Offered Load



Saturation knee ~ offered 0.015 / node / class | 168 nodes (84 RN + 84 HN), subnets=4, num_vcs=2

CHI Channel Mapping & Sweep Data

4 channels -> 4 independent physical networks

CHI channel	BookSim type	subnet	size	direction
REQ	read_request	0	1 flit	RN -> HN
DAT	read_reply	1	5 flit	HN -> RN
SNP	write_request	2	1 flit	-> RN
RSP	write_reply	3	1 flit	RN ->

Sweep results (latency in cycles, acc in flit/node/cycle)

offered	REQ/DAT lat	REQ/DAT acc	SNP/RSP lat	SNP/RSP acc
0.004	30.6	0.0278	28.1	0.0040
0.006	30.5	0.0398	28.3	0.0063
0.008	31.0	0.0536	28.6	0.0086
0.01	31.3	0.0660	28.4	0.0109
0.011	32.0	0.0716	29.0	0.0121
0.012	32.2	0.0779	29.3	0.0132
0.013	33.5	0.0839	29.9	0.0143
0.014	37.7	0.0875	38.2	0.0160
0.015	59.3	0.0903	105.3	0.0176
0.016	SAT	-	SAT	-

Highlighted row = last stable point before saturation (knee).